



Buffer Stock Saving with Heterogeneous Agents^{*}

An example of the Econ-ARK MyST template

Jane Economist¹  , and John Modeler^{1,2†}

¹Department of Economics, Johns Hopkins University , ²Econ-ARK; [†]Contributed Equally

Key points

- A buffer of wealth protects households against bad income draws.
- The endogenous grid method solves the model without root finding.
- Patience, returns and risk set the size of the buffer.

Abstract We solve a buffer stock saving model with the endogenous grid method and show how the Econ-ARK template typesets a working paper: title block, margin notes, equations, tables, code and author-year citations.

Non-technical summary Households save more when their income is uncertain. This paper shows how much more, and how quickly the extra saving can be computed.

Keywords consumption, precautionary saving, heterogeneous agents

JEL codes C63, D14, E21



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Code

github.com/econ-ark/econ-ark-myst

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Working paper

September 16, 2026

Cite as

Economist, Jane, and John Modeler. 2026. "Buffer Stock Saving with Heterogeneous Agents." *Econ-ARK Working Papers* 1 (3): 1-3.
<https://doi.org/10.5281/zenodo.0000000>
[arXiv:2609.00000](https://arxiv.org/abs/2609.00000)
[Zenodo archive](#)

Correspondence

Jane Economist
jane@example.org

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1 Introduction

Precautionary saving arises when income risk interacts with a convex marginal utility (Carroll 1997). The endogenous grid method (Carroll 2006) makes such models cheap to solve, because it replaces root finding with an inversion of the Euler equation.¹ Households hold a buffer of wealth against bad income draws. The size of that buffer depends on patience, the return on saving and the degree of risk.

2 Model

The consumer maximizes expected discounted utility subject to the budget constraint

$$v(m_t) = \max_{c_t} u(c_t) + \beta E_t[v(m_{t+1})], \quad m_{t+1} = R(m_t - c_t) + y_{t+1}. \quad (1)$$

The Euler equation implied by (1) is $u'(c_t) = \beta R E_t[u'(c_{t+1})]$, and with constant relative risk aversion $u(c) = c^{1-\rho}/(1-\rho)$.

^{*}Prepared for the Econ-ARK template documentation. Support from the Alfred P. Sloan Foundation is gratefully acknowledged. Computational Economics Grant (G-2026-00001). Jane Economist: Also a visiting scholar at Econ-ARK.

¹The method inverts the first-order condition on a grid of end-of-period assets rather than solving it on a grid of market resources.

2.1 Calibration

Table 1 Baseline calibration.

Parameter	Description	Value
β	Discount factor	0.96
R	Gross interest factor	1.03
ρ	Relative risk aversion	2.00

3 Results

Table 1 lists the parameters used in the solution. Proposition 1 describes the shape of the solution.

Proposition 1 (Concavity). *If income risk is present, the consumption function is strictly concave. Its slope falls toward the perfect foresight marginal propensity to consume as wealth grows.*

Proof. See Carroll (1997) for the argument under constant relative risk aversion. $\square$

```
from HARK.ConsumptionSaving.ConsIndShockModel import
IndShockConsumerType

agent = IndShockConsumerType()
agent.solve()
```

Appendix A derives the Euler equation.

Acknowledgments

We thank the Econ-ARK community for comments.

Data availability

All code needed to reproduce the results is in the repository linked in the margin.

Declaration of competing interest

The authors report no competing interests.

Declaration of generative AI use

The authors used a large language model to draft the example text of this template and reviewed every sentence before release.

References

Carroll, Christopher D. 1997. "Buffer-Stock Saving and the Life Cycle/Permanent Income Hypothesis." *The Quarterly Journal of Economics* 112 (1): 1–56. <https://doi.org/10.1162/003355397555109>.

Carroll, Christopher D. 2006. “The Method of Endogenous Gridpoints for Solving Dynamic Stochastic Optimization Problems.” *Economics Letters* 91 (3): 312–20. <https://doi.org/10.1016/j.econlet.2005.09.013>.

Appendix A Derivation of the Euler equation

The first-order condition for c_t in (1) sets $u'(c_t)$ equal to $\beta RE_t[v'(m_{t+1})]$. The envelope condition gives $v'(m_t) = u'(c_t)$, and substituting it yields the Euler equation.